

命题 1.

$$\log_a x \ll x^k \ll c^x \ll x^x (a > 0, k > 0, c > 1) (x \rightarrow +\infty)$$

命题 2.

$$(\ln x)^n \ll \frac{1}{x^p} (x \rightarrow 0)$$

其中 n 是正整数。

利用洛必达法则,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(\ln x)^n}{\frac{1}{x^p}} &= \lim_{x \rightarrow 0} \frac{(\ln x)^n}{x^{-p}} \\ &= \lim_{x \rightarrow 0} \frac{n(\ln x)^{n-1} \frac{1}{x}}{(-p)x^{-p-1}} \\ &= -\frac{n}{p} \cdot \lim_{x \rightarrow 0} \frac{(\ln x)^{n-1}}{x^{-p}} \\ &= -\frac{n}{p} \cdot \lim_{x \rightarrow 0} \frac{(n-1)(\ln x)^{n-2} \frac{1}{x}}{(-p)x^{-p-1}} \\ &= \left(-\frac{n}{p}\right) \left(-\frac{n-1}{p}\right) \cdot \lim_{x \rightarrow 0} \frac{(\ln x)^{n-2}}{x^{-p}} \\ &\vdots \\ &= \left(-\frac{n}{p}\right) \left(-\frac{n-1}{p}\right) \cdots \left(-\frac{2}{p}\right) \cdot \lim_{x \rightarrow 0} \frac{\ln x}{x^{-p}} \\ &= \left(-\frac{n}{p}\right) \left(-\frac{n-1}{p}\right) \cdots \left(-\frac{2}{p}\right) \left(-\frac{1}{p}\right) \cdot \lim_{x \rightarrow 0} \frac{1}{x^{-p}} \\ &= \left(-\frac{n}{p}\right) \left(-\frac{n-1}{p}\right) \cdots \left(-\frac{2}{p}\right) \left(-\frac{1}{p}\right) \cdot \lim_{x \rightarrow 0} x^p \\ &= 0 \end{aligned}$$